Guided practice

Circle applicable options; answer written questions with one or two sentences or key terms.

Refer to the historical image in A. Describe visible bacterial growth near the fungus, separating observation from an explanation of its cause.

Compare two claims: A, “This sample must kill all MRSA（耐甲氧西林金黃葡萄球菌）”; B, “Under the same conditions, the total clear-zone（清晰區） diameter around a disc with sample M is greater than around the carrier control.” Circle the more testable claim (A/B), and explain why clear-zone（清晰區） data can test it.

Hint: state the comparison and an observable outcome. Zone data cannot directly prove “all killed”.

Another context: compare unknown sample U with the same carrier control using a different bacterium. Write a hypothesis testable with external-zone presence and total diameter; do not make a new prediction about X/Y/Z.

What is C? Explain its comparison baseline and why the remaining conditions must be the same.

(Hint: The control group provides a baseline for comparison. It helps check whether changes in the experiment are caused by the independent variable rather than other external factors. Therefore, the control group does not receive the treatment being tested.)

Identify the variables

Tick all applicable boxes; more than one may be selected.

Independent variable（The factor you change）

- ☐ Disc size
- ☐ Sample in the disc
- ☐ Composition of the agar plate（瓊脂板）
- ☐ Visible clear zone（清晰區）and its total diameter
- ☐ Type of bacteria used
- ☐ Initial bacterial distribution
- ☐ Preset sample preparation

Dependent variable（The outcome you observe/measure）

- ☐ Disc size
- ☐ Sample in the disc
- ☐ Composition of the agar plate（瓊脂板）
- ☐ Visible clear zone（清晰區）and its total diameter
- ☐ Type of bacteria used
- ☐ Initial bacterial distribution
- ☐ Preset sample preparation

Controlled variables（The factors kept the same）

- ☐ Disc size
- ☐ Sample in the disc
- ☐ Composition of the agar plate（瓊脂板）
- ☐ Visible clear zone（清晰區）and its total diameter
- ☐ Type of bacteria used
- ☐ Initial bacterial distribution
- ☐ Preset sample preparation

What are the assumptions of this investigation?

An assumption is a condition we have not proved but take as true, to help make the experiment results valid.

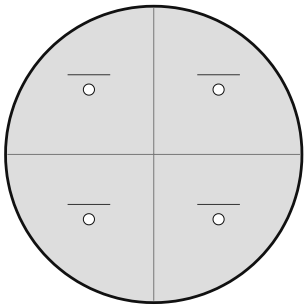
Select all applicable assumptions; you may choose more than one.

- ☐ Bacteria are initially distributed evenly on each agar plate（瓊脂板）.
- ☐ Equipment does not introduce contamination.
- ☐ All plates use the same simulated observation conditions.
- ☐ If no colonies（菌落） form near a disc, this means all bacteria at that location have died.

Complete labels and check fair comparison

Label the four discs X, Y, Z and C, once each. Explain why discs need space between them and from the edge. Label order does not affect credit.

You do not need to draw every detail carefully. A simple drawing with clear labels is enough.



Why are spacing and edge distance needed?

Circle flawed arrangements (more than one): A, “Pick up the next disc without re-sterilising the forceps”; B, “Use another bacterium on one plate, but combine all plates directly for comparison”; C, “Use the same carrier, bacteria and observation conditions”. Identify contamination or unfair comparison.

Plan A: remeasure the same plate three times. Plan B: independently prepare and observe three new plates once each. Which provides independent repeats, and how do they help check variation?

This lesson uses three plates. Why is “three” not the only correct number for every study? What feasibility limits matter when comparing three with more independent plates?

Diagram reading and procedure ordering (not practical performance)

These cards describe a simulated procedure. Order the letters logically; prepare three plates independently before the shared observation. Answering does not demonstrate sterilisation（滅菌／消毒）, spreading or incubation（培養） performance.

P. Observe all three prepared new plates under shared simulated conditions of 30°C and 24 hours.

Q. Spread vertically with a sterile（無菌的） spreader（塗布器） and rotate for even distribution.

R. Divide and label the plate bottom X/Y/Z/C.

S. Cover and invert; repeat the preparation independently on new plates.

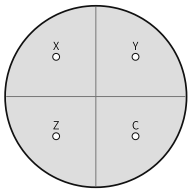
T. Before each disc, sterilise（滅菌／消毒） forceps with the alcohol beaker and lamp, then place it at the matching position.

U. Release diluted（已稀釋的） bacterial suspension（細菌懸液） at the top centre.

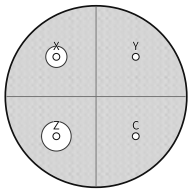
Circle errors (more than one): A, “Place discs before spreading evenly”; B, “Call remeasurement of one zone a new plate’s independent result”; C, “Retain differing readings from three independent plates”. State what should be checked.

Independent application: observe, measure and interpret • Plate 1

All plates and data are teaching simulations. Do not operate the VL or culture MRSA（耐甲氧西林金黃葡萄球菌） physically. Shared simulated conditions: 30°C and 24 hours.



Before simulated incubation（培養） (no visible growth result yet)

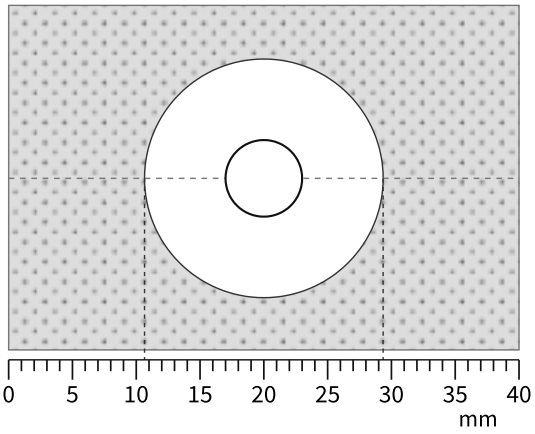


After simulated incubation（培養）: 30°C, 24 hours

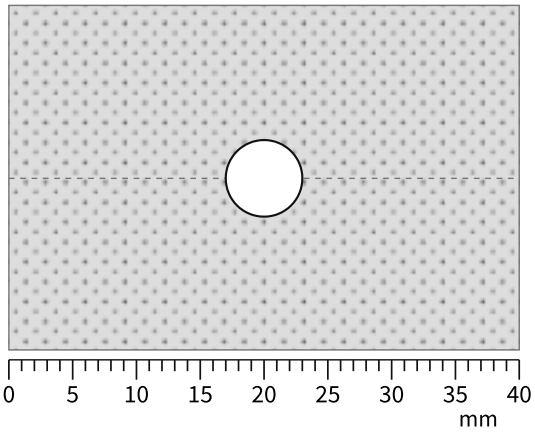
Read both ends on the illustrated mm scale and subtract for the total diameter. Use the labelled scale, not the physical length of the printed image.

When a zone extends beyond the disc, the total diameter includes the 6 mm disc. Without an external zone, record 0; the 6 mm disc alone is not a clear zone（清晰區）.

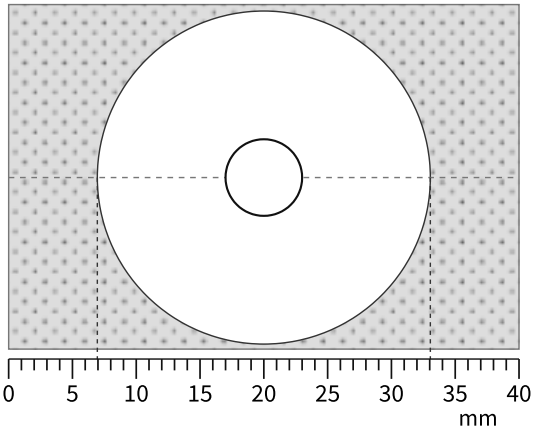
X



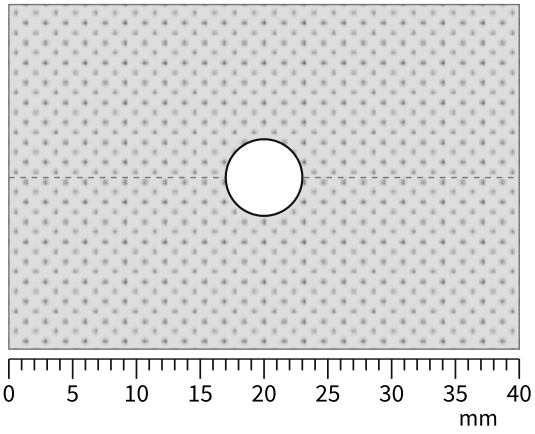
Y



Z



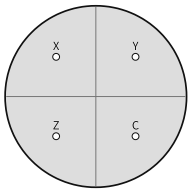
C (control)



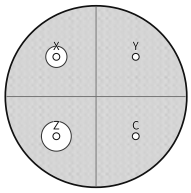
Sample	Visible zone outside the disc?	Total clear-zone（清晰區） diameter (mm)
X	☐ Yes ☐ No	
Y	☐ Yes ☐ No	
Z	☐ Yes ☐ No	
C (control)	☐ Yes ☐ No	

Independent application: observe, measure and interpret • Plate 2

All plates and data are teaching simulations. Do not operate the VL or culture MRSA（耐甲氧西林金黃葡萄球菌） physically. Shared simulated conditions: 30°C and 24 hours.



Before simulated incubation（培養） (no visible growth result yet)

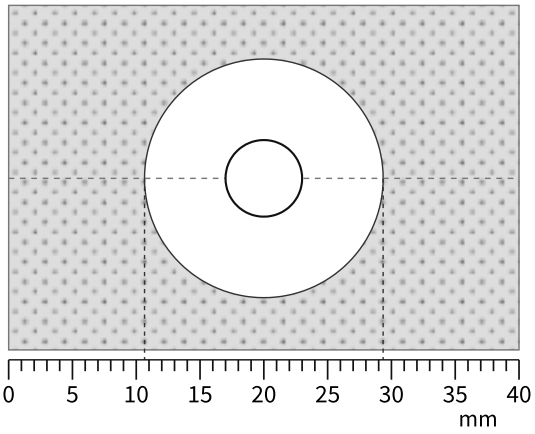


After simulated incubation（培養）: 30°C, 24 hours

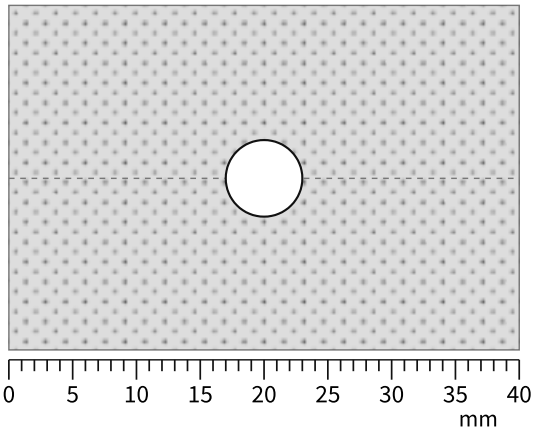
Read both ends on the illustrated mm scale and subtract for the total diameter. Use the labelled scale, not the physical length of the printed image.

When a zone extends beyond the disc, the total diameter includes the 6 mm disc. Without an external zone, record 0; the 6 mm disc alone is not a clear zone（清晰區）.

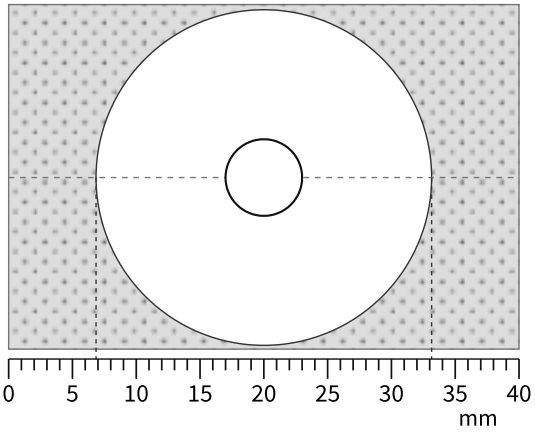
X



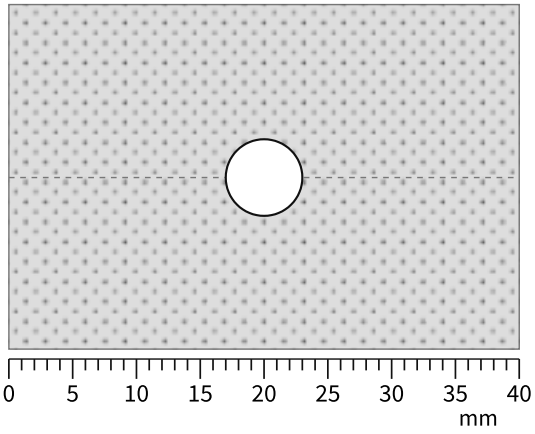
Y



Z



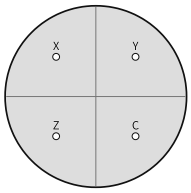
C (control)



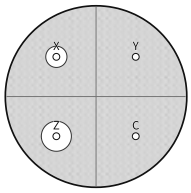
Sample	Visible zone outside the disc?	Total clear-zone（清晰區） diameter (mm)
X	☐ Yes ☐ No	
Y	☐ Yes ☐ No	
Z	☐ Yes ☐ No	
C (control)	☐ Yes ☐ No	

Independent application: observe, measure and interpret • Plate 3

All plates and data are teaching simulations. Do not operate the VL or culture MRSA（耐甲氧西林金黃葡萄球菌） physically. Shared simulated conditions: 30°C and 24 hours.



Before simulated incubation（培養） (no visible growth result yet)

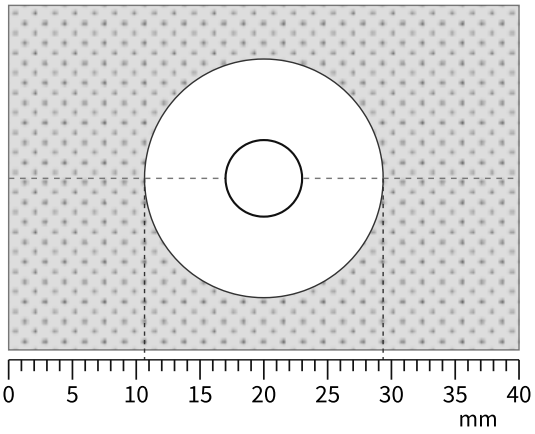


After simulated incubation（培養）: 30°C, 24 hours

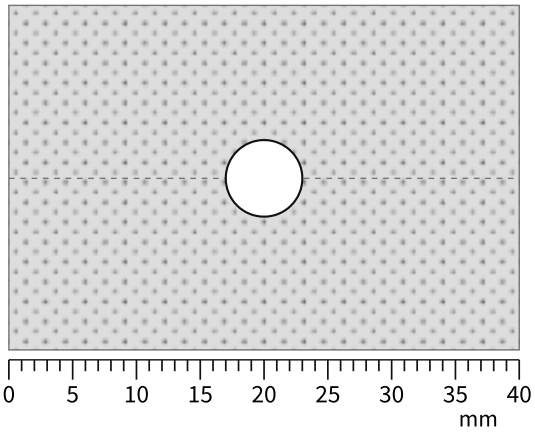
Read both ends on the illustrated mm scale and subtract for the total diameter. Use the labelled scale, not the physical length of the printed image.

When a zone extends beyond the disc, the total diameter includes the 6 mm disc. Without an external zone, record 0; the 6 mm disc alone is not a clear zone（清晰區）.

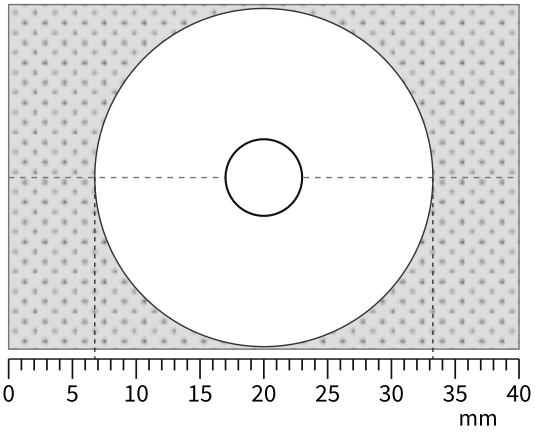
X



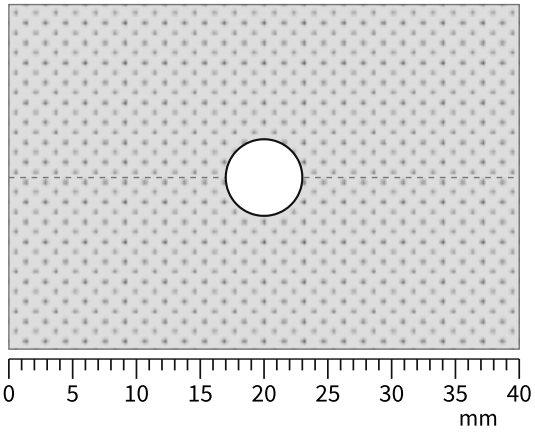
Y



Z



C (control)



Sample	Visible zone outside the disc?	Total clear-zone（清晰區） diameter (mm)
X	☐ Yes ☐ No	
Y	☐ Yes ☐ No	
Z	☐ Yes ☐ No	
C (control)	☐ Yes ☐ No	

Analyse and conclude

Use your three readings in B: mean = sum of three readings ÷ 3, to one decimal place.

Plate	X (mm)	Y (mm)	Z (mm)	C (control) (mm)
1				
2				
3				

Mean for X (mm) _____ Mean for Y (mm) _____

Mean for Z (mm) _____ Mean for C (control) (mm) _____

1. What does the appearance of a clear zone（清晰區） indicate?

○ A. All bacteria in the zone have died.
○ B. Under these test conditions, there is no visible bacterial growth in the zone, indicating inhibition（抑制）.
○ C. Bacteria in the zone resist every antibiotic（抗生素）.
○ D. The sample has eliminated viruses on the agar plate（瓊脂板）.

2. What does the diameter of a clear zone（清晰區） represent?

○ A. The exact number of bacteria remaining on the plate.
○ B. Its effectiveness in patients; the largest diameter identifies the best treatment.
○ C. The extent of the area with no visible bacterial growth under these conditions; diffusion and other factors also affect the diameter.
○ D. The antibiotic（抗生素） dose a patient should receive.

3. Draw a conclusion from your results:

Under these simulated conditions, samples （X ／ Y ／ Z ／ X and Y ／ X and Z ／ Y and Z ／ X, Y and Z ／ No samples） show a visible zone outside the disc.

Ranking of mean total clear-zone（清晰區） diameter under these conditions:

（X ／ Y ／ Z ／ C (control)） ＞ （X ／ Y ／ Z ／ C (control)） ＞ （X ／ Y ／ Z ／ C (control) ／ Not applicable） ＞ （X ／ Y ／ Z ／ C (control) ／ Not applicable）

Rank only samples with an external zone; select “Not applicable” for unused places. This is not a clinical effectiveness or treatment ranking.

4. Why do experiments need to be repeated?

○ A. To ensure that every experiment produces exactly identical absolute values, with no variation whatsoever.
○ B. To prolong the experiment so scientists have more to describe when writing their reports.
○ C. To reduce random error and assess the reproducibility（可重複性） of the results and the reliability（可靠性） of the data.
○ D. To make biological samples rapidly mutate and develop entirely new traits through repeated procedures.

5. If repeated results differ greatly, what should you do, and why?

○ A. Delete results that disagree with the original hypothesis so the results match.
○ B. Keep only the largest reading because it best represents inhibition（抑制）.
○ C. Calculate the mean immediately because averaging removes all procedural problems.
○ D. First check spreading, disc positions and plate conditions, then repeat as needed to identify causes of variation and improve the comparison.

6. When comparing the clear zones（清晰區） around C (control) and X, Y and Z, what is the main role of C?

○ A. Ensure that all discs have clear zones（清晰區） of the same diameter.
○ B. Identify the best antibiotic（抗生素） for a patient.
○ C. Provide an antibiotic（抗生素）-free comparison to help assess whether clear-zone（清晰區） formation is associated with the antibiotic（抗生素） sample.
○ D. Increase the amount of sample in the X, Y and Z discs to increase the accuracy（準確度） of the experimental results.

This model uses no clinical susceptibility（對抗生素的敏感性） breakpoints and cannot be generalised to all MRSA（耐甲氧西林金黃葡萄球菌）.

Concept application: selection of resistance（抗藥性）

Choose a growth-inhibiting（抑制） sample and cite readings. If a few bacteria already resist that antibiotic（抗生素）, how might continued use affect its effectiveness? Explain through survival, reproduction（繁殖）, inheritance（遺傳） and changing proportions.

Evidence to cite

Refer to your own twelve readings and means in B3–6. Use one effective sample; a short causal explanation is enough.

Complete B yourself and preserve your first answers before reading C. The examples in A are not the results for X/Y/Z/C.